CHAPTER 01 : COMPUTER FUNDAMENTAL

INTRODUCATION:-

Have you ever wondered how one machine can help us study, play games, watch videos, and communicate? This machine is called a computer. The word computer comes from "computare," which means to calculate.

In earlier times, people did calculations by hand and used tools like the Abacus. Later, Charles Babbage designed the Analytical Engine, and he is known as the Father of the Computer.

Today, computers are fast electronic machines used in many places like schools, homes, and banks. They work on instructions given by the user and follow the cycle of input, process, and output.

Example: When you type on the keyboard, the computer processes it and shows the result on the screen.

Defination:-

- A computer is an electronic device that takes data as input, processes it, and gives desired output.
- A computer is an electronic device that accepts data as input, processes it according to given instructions, and produces meaningful information as output.



📘 IPO Cycle (Input → Process → Output)

A computer works in a simple cycle called the IPO Cycle. IPO stands for Input, Process, and Output. This cycle explains how a computer performs any task.

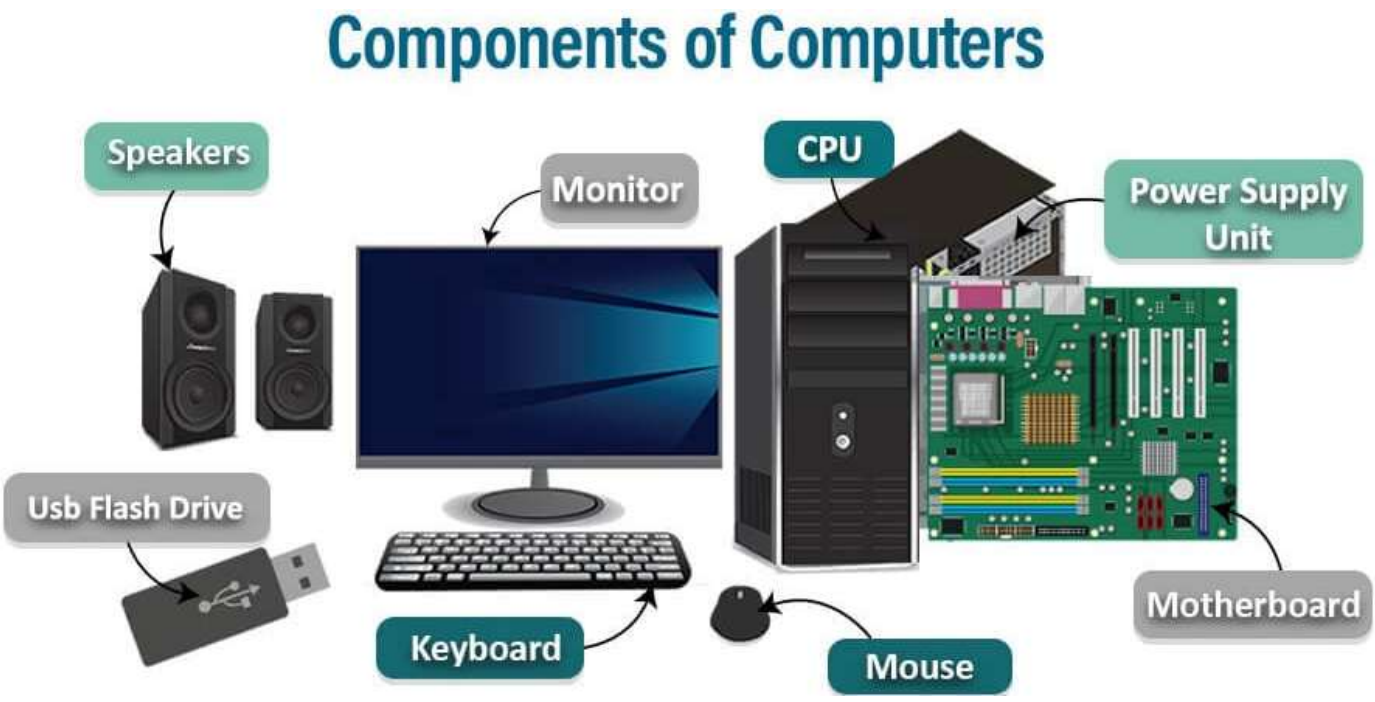


First, the computer takes data or instructions from the user. This step is called Input. Then, the computer works on that data using the CPU, which is called Processing. Finally, the computer gives the result to the user, which is called Output.

Example: When you type a sum using the keyboard (Input), the computer calculates it (Process) and shows the answer on the screen (Output).

Basic Components of Computer

A computer is made up of different parts that work together to perform tasks. These main parts are called the basic components of a computer.



The first component is Input Devices. These devices are used to enter data into the computer. For example, a keyboard is used for typing and a mouse is used for clicking and selecting.

The second component is the CPU (Central Processing Unit). It is known as the brain of the computer because it processes the data and controls all the activities of the computer.

The third component is Output Devices. These devices show the result of the work done by the computer. For example, a monitor displays the output on the screen and a printer prints it on paper.

Example: When you type using a keyboard (input), the CPU processes it, and the result is shown on the monitor (output).

Characteristics of a Computer

A computer has some special features that make it very useful in our daily life. These features are called the characteristics of a computer.

- 1. Speed

  A computer works very fast. It can perform many calculations in a few seconds.

- 2. Accuracy

  A computer gives correct results if the instructions are correct.

- 3. Storage

A computer can store a large amount of data such as files, images, and videos.

- 4. Diligence

A computer never gets tired. It can work continuously for a long time without making mistakes.

- 5. Versatility

A computer can perform different types of tasks like studying, drawing, playing games, and calculations.

Types of Computers

Computers are divided into different types based on their size, speed, and working capacity.



1. Supercomputer

- Supercomputers are the most powerful and fastest computers in the world.
- They can perform millions of calculations in a second and solve very complex problems.
- These computers are very large in size and expensive.
- They are mainly used in weather forecasting, space research, and scientific experiments.



2. Mainframe Computer

- Mainframe computers are large and powerful computers used by big organizations.
- They can store a huge amount of data and allow many users to work at the same time.
- They are very reliable and used for important tasks.
- These are commonly used in banks, railways, and government offices.

3. Mini Computer

- Mini computers are medium-sized computers with moderate speed and power.
- They are smaller than mainframe computers but can still support multiple users at once.
- They are less expensive and easier to manage.
- These are used in schools, small businesses, and offices.



4. Microcomputer (Personal Computer)



- Microcomputers are small and most commonly used computers.
- They are designed for personal use and are easy to operate.
- They are available in different forms like desktop, laptop, and tablets.

- These are used at home, in schools, and in offices for daily tasks like studying, typing, and entertainment.

Basic Terms of Computer

To understand how a computer works, it is important to learn some basic terms used in computing. These terms help us understand the functioning of a computer in a better way.

1. Data

Data refers to raw facts and figures that are given to the computer for processing. It may include numbers, text, or symbols.
Example: 25, “Rahul”, A+

2. Information

Information is the processed form of data that is meaningful and useful.
Example: Student result or report card

3. Input

Input is the data or instructions entered into the computer through input devices.
Example: Typing using a keyboard

4. Output

Output is the result produced by the computer after processing the input data.
Example: Result displayed on the monitor

5. Processing

Processing is the operation performed by the computer on input data to convert it into useful information.
Example: Performing calculations

6. Storage

Storage refers to the process of saving data and information for future use.
Example: Saving files in a computer

7. CPU (Central Processing Unit)

The CPU is the main part of the computer that controls all operations and processes data. It is also known as the **brain of the computer**.

Exercise

A. Multiple Choice Questions (MCQ)

1. A computer is a _____ machine.
   a) Manual
   b) Electronic
   c) Mechanical
   d) Natural

2. CPU stands for _____.
   a) Computer Process Unit
   b) Central Print Unit
   c) Central Processing Unit
   d) Control Processing Unit

3. Which is an input device?
   a) Keyboard
   b) Monitor
   c) Speaker
   d) Printer

4. Which is the brain of the computer?
   a) Mouse
   b) CPU
   c) Keyboard
   d) Monitor

5. IPO stands for _____.
   a) Input Print Output
   b) Input Program Output
   c) Internal Process Output
   d) Input Process Output

6. Which of the following is an output device?
   a) Keyboard
   b) Mouse
   c) Monitor
   d) Scanner

7. Which device is used to move the cursor on screen?
   a) Keyboard
   b) Mouse
   c) CPU
   d) Printer

8. Which of the following is the fastest computer?
   a) Laptop
   b) Desktop
   c) Supercomputer
   d) Tablet

B. Fill in the Blanks

1. A computer is an _____ machine.
2. CPU is the _____ of the computer.
3. IPO stands for _____, _____, _____.
4. _____ is used to type data into the computer.
5. Monitor is an _____ device.

🔹 C. True / False

1. A computer can think on its own. (____)
2. CPU controls all the operations of a computer. (____)
3. Keyboard is an output device. (____)
4. Monitor is an output device. (____)
5. Supercomputer is the fastest computer. (____)

🔹 D. Short Answer Questions (Limited)

1. What is a computer?
2. What is IPO cycle?
3. Name two input devices.
4. What is CPU?
5. What is a supercomputer?
6. Write characteristics of a computer.

🔹 E. Activity

👉 Draw and label the IPO cycle:
Input → Process → Output

ANSWER KEY

Q1 MCQ

1. b
2. c
3. a
4. b
5. d
6. c
7. b
8. c

Q2 FIB

1. Electronic
2. Brain
3. Input, Process, Output
4. Keyboard
5. Output

Q3 TRUE OF FALSE

1. False
2. True
3. False
4. True
5. True

1. **What is a computer?**
   A computer is an electronic machine that accepts data as input, processes it according to given instructions, and produces meaningful output. It can perform calculations, store information, and help in many tasks.

2. **What is IPO cycle?**
   IPO cycle stands for Input, Process, and Output. It explains how a computer works. First, data is entered using input devices (Input), then the CPU processes the data (Process), and finally the result is shown through output devices (Output).

3. **Name two input devices.**
   Keyboard and Mouse are two common input devices. A keyboard is used to type data, while a mouse is used to control the cursor and select items on the screen.

4. **What is CPU?**
   CPU (Central Processing Unit) is the main part of the computer. It is called the brain of the computer because it controls all operations and processes data to produce results.

5. **What is a supercomputer?**
   A supercomputer is the fastest and most powerful type of computer. It is used for complex calculations like weather forecasting, scientific research, and space studies.